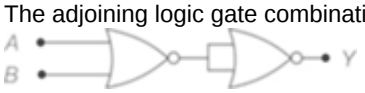
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MM : 720 NBTS pre **Time : 44 Min.**

physics -12

1. The adjoining logic gate combination is equivalent to



(1) OR

(2) NOR

(3) AND

(4) NAND

2. A nucleus of mass number A has mass defect Δm. What is the formula for the binding energy per nucleon of this nucleus?

(1) $\frac{\Delta mc^2}{A}$

(2) $\frac{\Delta m^2 c}{A}$

(3) $\frac{\Delta mc}{A}$

(4) $(\Delta mc^2)A$

3. If the emitted radiation falls in the microwave region, the device is termed as

(1) Laser

(2) Maser

(3) Both (1) & (2)

(4) None of these

4. The energy released after complete consumption of 1 g of matter is

(1) 9×10^{16} J

(2) 2.5×10^{7} kWh

(3) 1.6×10^{16} eV

(4) Both (1) and (2)

5. A particle of mass *m* moves in a circular orbit (under the influence of a central force) with potential energy $U(r) = \frac{cr^3}{3}$, where *c* is a positive constant and *r* is the radial distance from the origin. Assuming the Bohr's quantisation rule to be valid, the total energy of the particle will be

(1) $\frac{1}{3}\left(\frac{n^{10}h^{10}}{1024\pi^{10}m^5c^2}\right)^{1/3}$

(2) $\frac{1}{3}\left(\frac{n^6h^6c^2}{64\pi^6m^3}\right)^{1/5}$

(3) $\frac{5}{6}\left(\frac{n^6h^6c^2}{64\pi^6m^3}\right)^{1/5}$

(4) $\frac{5}{6}\left(\frac{n^{10}h^{10}}{1024\pi^{10}m^5c^2}\right)^{1/3}$

6. A intrinsic semiconductor have equal electron and hole concentration of 1.5×10^{10} cm^{-3}. Doping by phosphorus, increases electron concentration to 2.25×10^{22} m^{-3}. The hole concentration in doped semiconductor is

(1) 10^{10} m^{-3}

(2) 10^{9} m^{-3}

(3) 10^{11} m^{-3}

(4) 5×10^{9} m^{-3}

7. If radius of $_{13}Al^{27}$ nucleus is estimated to be 3.6 fermi, then radius of $_{52}Te^{125}$ nucleus is nearly

(1) 4 fermi

(2) 6 fermi

(3) 8 fermi

(4) 3.2 fermi

8. The radius of $_{32}X$ nuclide is twice that of $_{4}Be^{9}$. The number of neutrons in *X* is

(1) 40

(2) 72

(3) 32

(4) 36

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9. A half wave rectifier circuit uses diode with forward resistance 20 Ω and has load resistance of 50 Ω. If 10 : 1 turn ratio transformer is used and primary A. C voltage is 210 V, 50 Hz, then the peak value of output current is

(1) $\frac{3}{\sqrt{2}}A$

(2) $\frac{3}{5}A$

(3) $\frac{3}{5\sqrt{2}}A$

(4) $\frac{3\sqrt{2}}{5}A$

10. The wavelength of radiation emitted, when an electron jumps from the third to second orbit of hydrogen atom is λ. The wavelength of radiation emitted for the electron jumps from the fourth to second orbit of hydrogen atom will be

(1) $\frac{27\lambda}{20}$

(2) $\frac{20\lambda}{27}$

(3) $\frac{4\lambda}{5}$

(4) $\frac{5\lambda}{4}$

11. Which of the following is used in forward bias for their operation?

(1) LED

(2) Photodiode

(3) Solar cell

(4) Zener diode

12. Reverse bias voltage applied to a junction diode

(1) Decreases the potential barrier

(2) Increases the potential barrier

(3) Increases the majority carrier current

(4) Decreases the minority carrier current

13. If the speed of an electron in 2nd energy level of H-like atom is *v*, then speed of electron in its 4th energy level is

(1) $\frac{v}{2}$

(2) $\frac{v}{4}$

(3) $\frac{v}{3}$

(4) *v*

14. The nuclides with same neutron number N but different atomic number Z, are called

(1) Isobars

(2) Isotones

(3) Isotopes

(4) Both (1) and (3)

15. The correct statement about radioactive decay among the following is

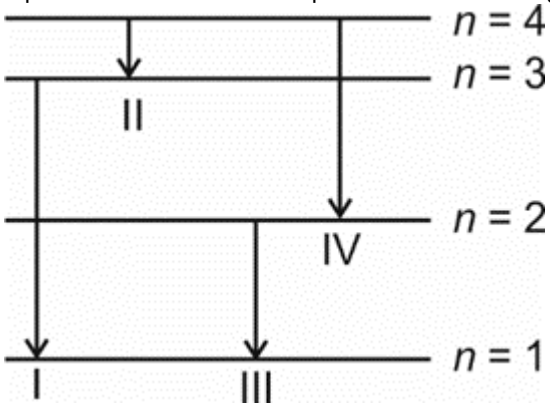
(1) In a γ-decay, the element does not change

(2) In an α-decay, the helium nucleus is emitted

(3) In a β^+ decay, positron is emitted from the nucleus

(4) All of these

16. The diagram shows the energy levels for an electron in a certain atom. Which transition among the following represents the emission of a photon with least energy



(1) II

(2) III

(3) I

(4) IV

17. A neutron strikes a $_{92}U^{235}$ nucleus and as a result $_{36}Kr^{93}$ and $_{56}Ba^{140}$ are produced with

(1) α-particle

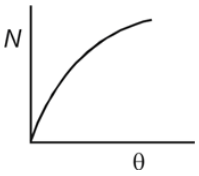
(2) 1-neutron

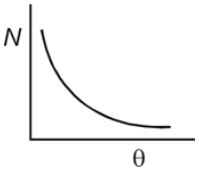
(3) 3-neutron

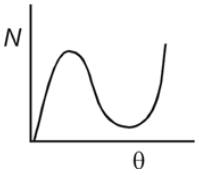
(4) 2-β-particle

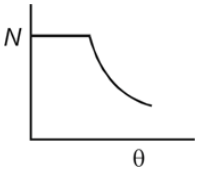
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18. Which of the following may be representing graph between number of scattered particles detected (N) and scattering angle (θ) in Rutherford's experiment?

(1) 

(2) 

(3) 

(4) 

19. When a p-n junction is forward biased then the major fraction of current through the junction is due to

(1) Diffusion of charges

(2) Drift of charges

(3) Both drift and diffusion of charge

(4) Neither drift nor diffusion

20. For a hydrogen atom, some of the parameters associated with electron given in column-I and their relation with principal quantum number n in column-II. Match column-I with column-II.

	Column-I		**Column-II**
A	Radius of orbit (r)	(i)	$\frac{1}{n}$
B.	Velocity (v)	(ii)	n^2
C.	Time period (T)	(iii)	$\frac{1}{n^5}$
D.	Magnetic field produced (B) at centre	(iv)	n^3

(1) A(ii), B(i), C(iv), D(iii)

(2) A(i), B(ii), C(iii), D(iv)

(3) A(i), B(iii), C(ii), D(iv)

(4) A(iv), B(i), C(ii), D(iii)

21. Isotones are nuclides having the same number of

(1) Protons

(2) Neutrons

(3) Electrons

(4) Both (1) and (2)

22. The ratio of shortest wavelengths of hydrogen atom in Lyman, Balmer and Paschen series are in the ratio of

(1) 1 : 2 : 3

(2) 1 : 4 : 9

(3) 9 : 1 : 3

(4) 4 : 12 : 17

23. Consider the following statements:
a. The density of the nuclear matter is independent of the size of the nucleus.
b. Fusion of light nuclei to form heavier nuclei is possible at room temperature.
c. Electron and positron are particle-antiparticle pair.
The correct statement(s) is/are

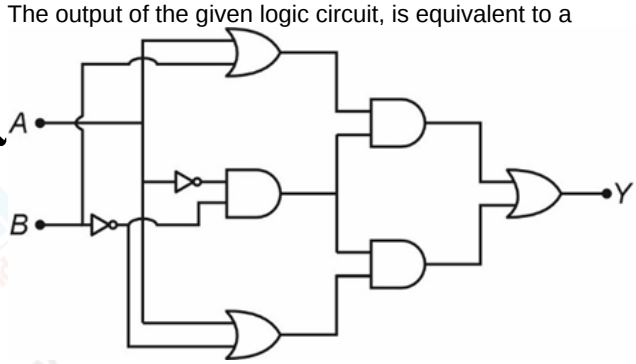
(1) Both a and c

(2) Both a and b

(3) Both b and c

(4) Only b

24. The output of the given logic circuit, is equivalent to a



(1) OR gate

(2) AND gate

(3) NOR gate

(4) NAND gate

25. The binding energy per nucleon for deuteron ($_1H^2$) and helium ($_2He^4$) are 1.1 MeV and 7.0 MeV respectively. The energy released when two deuterons fuse to form a helium nucleus $_2He^4$ is

(1) 26.3 MeV

(2) 23.6 MeV

(3) 26.3 eV

(4) 23.6 eV

26. Ratio of nuclear radius of ^{27}Al and that of ^{125}Fe is

(1) 1 : 1

(2) 1 : 3

(3) 3 : 5

(4) 27 : 125

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27. A nucleus of mass *M* emits γ- rays of energy *E*. The recoil energy of the nucleus is

(1) *E*

(2) Mc^2

(3) $\frac{E^2}{2Mc^2}$

(4) $\frac{1}{2}Mc^2$

28. The main function of a moderator in a nuclear reactor is to

(1) Accelerate neutrons

(2) Slow down fast neutrons

(3) Absorb fast neutrons

(4) Absorb the uranium

29. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): In a spontaneous fission reaction, when a heavy nucleus is split into two medium sized ones, then there is release of energy.
Reason (R): The original nucleus has more binding energy per nucleon than each of the two new product nuclei.
In the light of the above statements, choose the correct answer from the options given below:

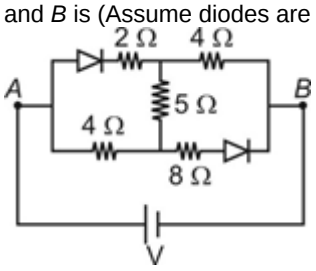
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

30. The equivalent resistance of the circuit across points *A* and *B* is (Assume diodes are ideal)



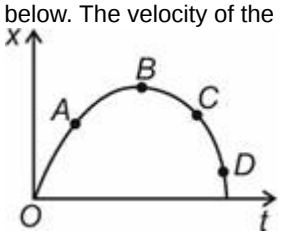
(1) 4 Ω only

(2) 13 Ω only

(3) 15 Ω only

(4) Either 4 Ω or 13 Ω

31. The position *x* versus time *t* graph for a car is shown below. The velocity of the car is zero at point



(1) *A*

(2) *B*

(3) *C*

(4) *D*

32. A : Average velocity is a vector quantity.
R : Average speed ≥ |average velocity|.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

33. A bullet fired in a fixed wooden block loses half of its velocity after penetrating 6 cm. Assuming the bullet experiences constant retardation, how much more distance will it travel before coming to a complete stop?

(1) 2 cm

(2) 6 cm

(3) 9 cm

(4) 4 cm

34. The relation between time (*t*) and position (*x*) is $t = 3x^2 + 2x$. The relation between magnitude of velocity and magnitude of acceleration is

(1) $a \propto v$

(2) $a \propto v^3$

(3) $a \propto v^2$

(4) $a \propto \frac{1}{v}$

35. A car covers first half of the total distance with speed 20 km/h and second half of total distance with speed 60 km/h. The average speed of the car for the entire trip is

(1) 30 km/h

(2) 40 km/h

(3) 15 km/h

(4) 20 km/h

36. A body start from rest and moves with a constant acceleration of 5 m/s^2. Its instantaneous speed at the end of 10 s is

(1) 50 m/s

(2) 5 m/s

(3) 25 m/s

(4) Zero

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37. **Assertion (A) :** A body moving with increasing acceleration may have decreasing speed.
Reason (R) : When acceleration of body is in opposite direction to velocity, speed will decrease.

(1) If both Assertion & Reason are true and the reason is the correct explanation of the assertion, then mark **(1)**

(2) If both Assertion & Reason are true but the reason is not the correct explanation of the assertion, then mark **(2)**

(3) If Assertion is true statement but Reason is false, then mark **(3)**

(4) If both Assertion and Reason are false statements, then mark **(4)**

38. For a particle undergoing uniform motion along a straight line, the correct statement is:
(A) Its position-time graph is a straight line.
(B) Its velocity-time graph is a straight line.
(C) Its acceleration-time graph is a straight line.
Choose the correct answer among the options given below.

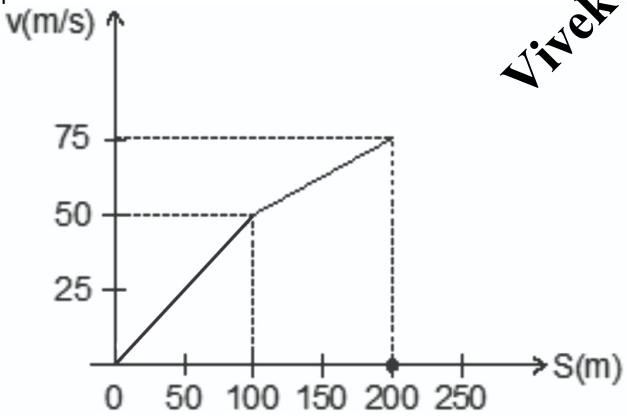
(1) (A) only

(2) (B) Only

(3) (B) and (C) only

(4) All (A), (B) and (C)

39. The velocity-position ($v - s$) graph for a particle moving on a straight runway is as shown in figure. The acceleration of particle at S = 150 m is



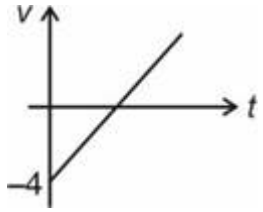
(1) $\frac{125}{8}$ m/s^2

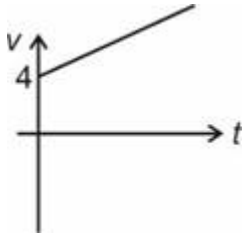
(2) $\frac{125}{4}$ m/s^2

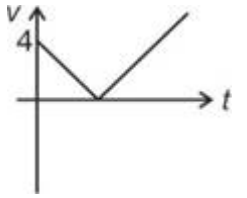
(3) $\frac{250}{4}$ m/s^2

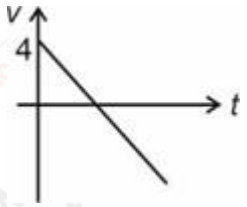
(4) 60 m/s^2

40. A particle moves along x-axis in such a way that its x co-ordinate varies with time t according to the equation $x = 5 - 4t + t^2$. The speed v of the particle will vary with time t as (All quantities are in SI units)

(1)


(2)


(3)


(4)


41. Ratio of distance covered by a body to the magnitude of its displacement may be

(1) Less than 1

(2) Equal to 1

(3) Greater than 1

(4) Both (2) and (3)

42. Velocity of a body moving with uniform acceleration of 2 m/s^2 is changed by 20 m/s in certain time. Average velocity of body during this time is 20 m/s. Distance covered by it during this time will be

(1) 400 m

(2) 200 m

(3) 100 m

(4) 800 m

43. A particle is moving in a straight line under constant acceleration. If the motion starts from rest then the ratio of displacement in 5 second to that in the 5th second

(1) $\frac{16}{25}$

(2) $\frac{9}{25}$

(3) $\frac{25}{16}$

(4) $\frac{25}{9}$

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44. A pebble is released into a well of depth *h*. The splash is heard after time *t* second of release. If *v* is velocity of sound in air, then

(1) $t = \frac{gh^2}{v}$

(2) $t = \sqrt{\frac{2h}{g}} + \frac{h}{v}$

(3) $\sqrt{\frac{h}{g}} + \frac{2h}{v}$

(4) $\frac{2h}{v}$

45. A boy takes 4 h to cover certain distance along the flow and takes 6 h to cover the same distance opposite to flow. In how much time, he will cross this distance in still water?

(1) 4.5 h

(2) 5 h

(3) 2.4 h

(4) 4.8 h

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Chemistry

46. Given below are two statements. One is labelled as **assertion (A)** and other is labelled as **reason (R)**.
Assertion (A): In strongly acidic medium, aniline becomes more reactive towards electrophilic reagent.
Reason (R): $-NH_2$ of aniline gets protonated in acidic medium and lone pair of electron is no longer available for resonance.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

47. Consider the following reaction sequence
(i) Metal phosphide + H_2O → A(g) + Metal Hydroxide
(ii) A(g) + $CuSO_4$(aq) → B + C (dibasic acid)
Select the correct statement among the following.

(1) Gas A is yellow in colour

(2) Gas A is significantly soluble in water

(3) Concentrated acid C oxidises sucrose to CO_2

(4) Gas A explodes in contact with trace amount of HNO_3

48. Match the chemical reactions given in column-I with their respective catalysts given in column-II and choose the most appropriate match from the options given below.

	Column-I		**Column-II**
a.	$2SO_2 + O_2 \rightarrow 2SO_3$	(i)	Fe
b.	$4HCl + O_2 \rightarrow 2Cl_2 + 2H_2O$	(ii)	Pt/Rh gauge
c.	$4\,NH_3 + 5O_2 \xrightarrow[9\ bar]{500\,K} 4NO + 6H_2O$	(iii)	$CuCl_2$
d.	$N_2 + 3H_2 \overset{773\,K}{\rightleftharpoons} 2\,NH_3$	(iv)	V_2O_5

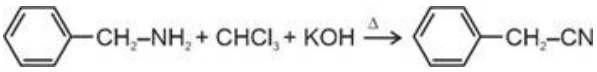
(1) a-(ii), b-(iii), c-(iv), d-(i)

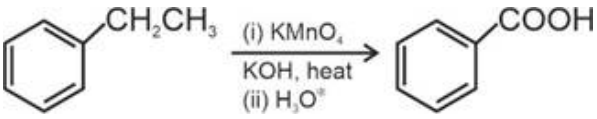
(2) a-(iv), b-(iii), c-(i), d-(ii)

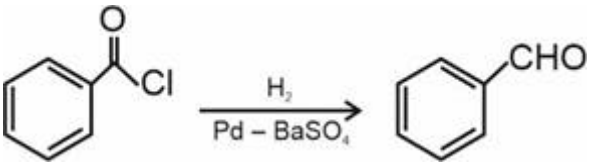
(3) a-(iv), b-(iii), c-(ii), d-(i)

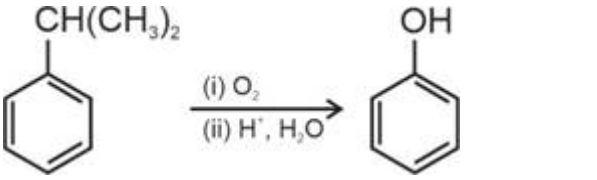
(4) a-(iii), b-(iv), c-(i), d-(ii)

49. Select the incorrect reaction among the following.

(1) 

(2) 

(3) 

(4) 

50. Match compound in list-I with their pK_b in list-II

	List-I		**List-II**
a.	Methanamine	(i)	4.22
b.	N-Methylmethanamine	(ii)	3.38
c.	N, N-Dimethylmethanamine	(iii)	3.29
d.	Ethanamine	(iv)	3.27

Correct match is

(1) a(iii), b(iv), c(i), d(ii)

(2) a(i), b(iv), c(iii), d(ii)

(3) a(iii), b(i), c(iv), d(ii)

(4) a(ii), b(iv), c(i), d(iii)

51. Number of chiral carbons in cyclic structure of α-D-fructose is

(1) 3

(2) 4

(3) 5

(4) 6

52. In qualitative analysis when H_2S gas is passed through an aqueous solution of salt acidified with dilute HCl, a black ppt. is obtained. On boiling the precipitate with dilute HNO_3 it forms a solution of blue colour. Addition of excess of aqueous solution of NH_3 to this solution gives

(1) Deep blue ppt of $Cu(OH)_2$

(2) Deep blue solution of $[Cu(NH_3)_4]^{2+}$

(3) Deep blue solution of $Cu(NO_3)_2$

(4) Deep blue solution of $Cu(OH)_2 \cdot Cu(NO_3)_2$

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53. Given below are the two statements. One is labelled as assertion (A) and other is labelled as reason (R).?
Assertion (A): Ammonolysis of excess of alkyl halide at 373 K produces a mixture of 1° amine, 2° amine, 3° amine and quaternary ammonium salt.
Reason (R): NH_3 firstly reacts with RX to produce 1° amine. The 1° amine thus obtained behaves as a nucleophile and can further react with alkyl halide to form 2° and 3° amines, and finally quaternary ammonium salt.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) (A) is true but (R) is false

(3) (A) is false but (R) is true

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

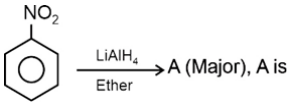
54. The correct order of boiling point of hydrides of group 15 elements is

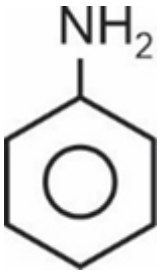
(1) $NH_3 > PH_3 > AsH_3 > SbH_3$

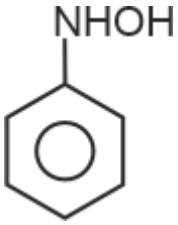
(2) $SbH_3 > AsH_3 > PH_3 > NH_3$

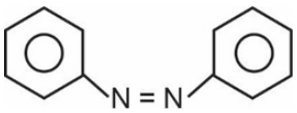
(3) $SbH_3 > NH_3 > AsH_3 > PH_3$

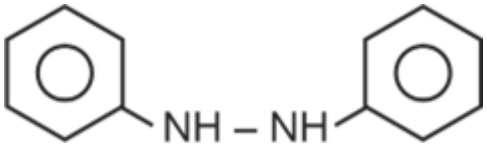
(4) $NH_3 > SbH_3 > PH_3 > AsH_3$

55. 

(1) 

(2) 

(3) 

(4) 

56. Consider the following statements about colloids
(a) Generally, size of colloidal particles ranges from 1–1000 nm
(b) Starch sol is an example of lyophilic sol.
(c) Stability of lyophobic sol is due to charge on colloidal particles.
The correct statements are

(1) (a) and (b) only

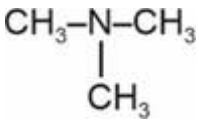
(2) (b) and (c) only

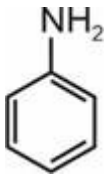
(3) (a) and (c) only

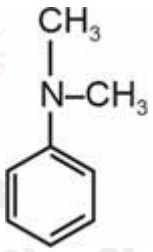
(4) (a), (b) and (c)

57. Which of the following amines will give the positive carbylamine test?

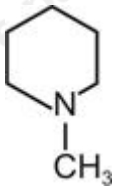
(1) CH_3NHCH_3

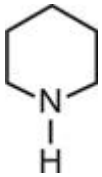
(2) 

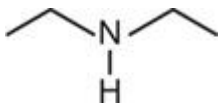
(3) 

(4) 

58. The amine which will not react with Hinsberg's reagent is

(1) 

(2) 

(3) 

(4) 

59. Correct order of the boiling point of the given compounds is

(1) $BiH_3 > SbH_3 > NH_3 > AsH_3 > PH_3$

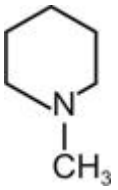
(2) $NH_3 > BiH_3 > SbH_3 > AsH_3 > PH_3$

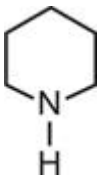
(3) $NH_3 > BiH_3 > SbH_3 > PH_3 > AsH_3$

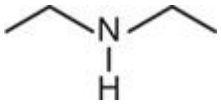
(4) $BiH_3 > NH_3 > SbH_3 > AsH_3 > PH_3$

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60. The amine which will not react with Hinsberg's reagent is .

(1) 

(2) 

(3) 

(4) 

61. Amino acids are known to be building blocks of

(1) Proteins

(2) Nucleic acids

(3) Carbohydrates

(4) Vitamins

62. Given below are two statements one is labelled as **Assertion (A)** and other is labelled as **Reason (R)**
Assertion (A) Starch is used as an indicator in iodometric titrations.
Reason (R): Starch forms iodo starch complex with iodine, which is red in colour
In light of above statements, choose the most appropriate answer from the options given below.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct and (R) is not the correct explanation of (A)

(3) (A) is correct, but (R) is incorrect

(4) (A) is incorrect, but (R) is correct

63. Consider the following reaction.
$XeF_6 + 2H_2O \rightarrow X + 4HF$
The hybridisation of central atom of X and shape of compound X respectively, are

(1) sp^3d, linear

(2) sp^3d, T-shape

(3) sp^3d^2, square pyramidal

(4) sp^3d, see-saw

64. Among the following, the correct order of acidic strength is

(1) $HClO_3 < HClO_4 < HClO_2 < HClO$

(2) $HClO_4 < HClO_2 < HClO < HClO_3$

(3) $HClO < HClO_2 < HClO_3 < HClO_4$

(4) $HClO_2 < HClO < HClO_3 < HClO_4$

65. Non-essential amino acid among the following is

(1) Valine

(2) Leucine

(3) Glycine

(4) Lysine

66. Convulsions and scurvy is caused by deficiency of X and Y respectively. X and Y respectively are

(1) Vitamin A and Vitamin C

(2) Vitamin B_6 and vitamin C

(3) Vitamin B_1 and vitamin B_2

(4) Vitamin C and vitamin B_1

67. Deficiency of which of the vitamin causes rickets?

(1) Vitamin-D

(2) Vitamin-C

(3) Vitamin-B_{12}

(4) Vitamin-E

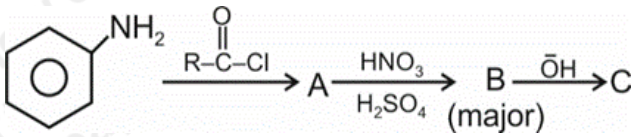
68. Methyl isocyanate, the poisonous gas which was responsible for Bhopal tragedy in 1984, on alkaline hydrolysis gives

(1) Methylamine

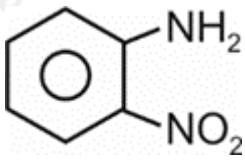
(2) Methyl carbamic acid

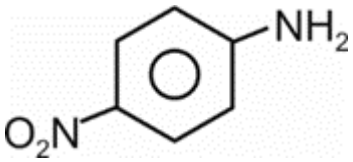
(3) Ethyl amine

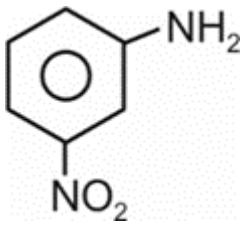
(4) Methyl cyanide

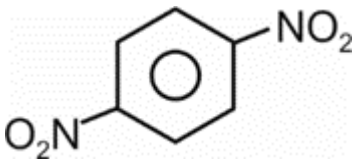
69. 

The product C is

(1) 

(2) 

(3) 

(4) 

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70. Which of the following gives a black precipitate?

(1) $PbCrO_4$

(2) AgBr

(3) $PbSO_4$

(4) PbS

71. Acetylation of glucose with acetic anhydride gives

(1) Glucose hexaacetate

(2) Glucose pentaacetate

(3) Glucose butaacetate

(4) Glucose diacetate

72. Alcoholic compounds on reaction with ceric ammonium nitrate gives a red colouration due to the formation of

(1) $[Ce(NO_3)_4\ (ROH)_3]$

(2) $[Ce(NH_4)_4(ROH)_3](NO_3)_3$

(3) $[Ce(H_2O)_4(ROH)_2](NO_3)_2$

(4) $[Ce(ROH)_6](NO_3)_2$

73. In the following questions, a statement of assertion (A) is followed by a statement of reason (R)
Assertion (A) : Nitration of aniline at low temperature gives 47% m-nitroaniline.
Reason (R) : In acidic medium, NH_2 group is converted into $-\overset{+}{N}H_3$ group which is m-directing.
In the light of above two statements choose the **correct** option.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

74. Negative electron gain enthalpy of halogens follows the order

(1) I > Br > Cl > F

(2) F > Cl > Br > I

(3) Cl > F > Br > I

(4) Cl > Br > I > F

75. Find the incorrect match.

	Column I		**Column II**
(1)	Keratin	–	Fibre-like structure
(2)	Insulin	–	Spherical shape
(3)	Albumin	–	Insoluble in water
(4)	Myosin	–	Hydrogen and disulphide bonds

(1) (1)

(2) (2)

(3) (3)

(4) (4)

76. Consider the following statements
(a) d_{z^2} and $d_{x^2-y^2}$ have electron density along the axes.
(b) The maximum number of 3p electrons in an atom having spin quantum value of $-\frac{1}{2}$ is three.
(c) Shape of orbital is determined by Azimuthal quantum number.
(d) K^+, Ca^{2+} and S^{2-} are isoelectronic species.
(e) The number of protons, neutrons and electrons in $^{23}_{11}Na$, respectively are 11, 12 and 11.
The correct statements are

(1) (a), (b) and (c) only

(2) (a) and (e) only

(3) (c), (d) and (e) only

(4) (a), (b), (c), (d) and (e)

77. The energy of an electron in a Bohr's orbit of hydrogen atom is – 1.51 eV. The radius of orbit is

(1) 4.76 Å

(2) 2.12 Å

(3) 1.06 Å

(4) 1.59 Å

78. 0.25 gram atom of an element weighs 25 g. The atomic mass of the elements is

(1) 50

(2) 100

(3) 200

(4) 25

79. If 30 ml of H_2 reacts with 20 ml of O_2 to form water. What is left at the end of the reaction?

(1) 5 ml of H_2

(2) 10 ml of H_2

(3) 10 ml of O_2

(4) 5 ml of O_2

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80. Given below are two statements

Statement I: The radius of the second orbit of He^+ is equal to that of the first orbit of hydrogen.
Statement II: The radius of an orbit in hydrogen like species is directly proportional to n and inversely proportional to Z.
In the light of above statements, choose the most appropriate answer from the option given below.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

81. Number of unpaired electrons present in Fe^{2+} ion is (Atomic number of Fe = 26)

(1) 4

(2) 3

(3) 5

(4) 2

82. Number of significant figures in the sum of 3.26 and 0.023 is

(1) 1

(2) 2

(3) 3

(4) 4

83. Which one of the following is not correct according to postulates of Bohr's atomic model?

(1) Energy of the electron only changes when it makes a transition from one energy level to the other

(2) An electron around the nucleus can revolve only in certain circular path of fixed radii and energy

(3) Angular momentum of an electron is quantized

(4) An electron moves in an orbit with continuous loss of energy

84. How many maximum electrons can be accommodated in an orbital for which n = 4 and l = 2?

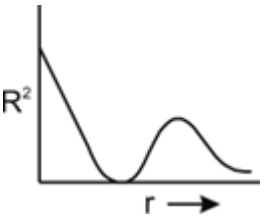
(1) 5

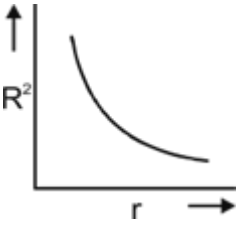
(2) 10

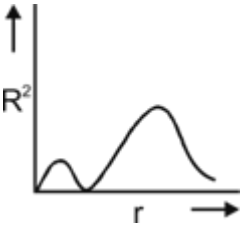
(3) 2

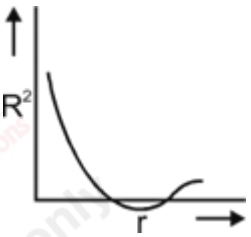
(4) 6

85. The probability density curve for 2s electron appears like

(1) 

(2) 

(3) 

(4) 

86. A solution containing 2 g per 100 mL of urea (molecular mass = 60 g mol^{-1}) is isotonic with a 5% (w/v) solution of a non-volatile solute. The molecular mass of this non-volatile solute is

(1) 175 g mol^{-1}

(2) 200 g mol^{-1}

(3) 350 g mol^{-1}

(4) 150 g mol^{-1}

87. The number of electrons present in 36 mL of water at 25°C is (Density of water = 1 g mL^{-1})

(1) 12×10^{23}

(2) 12×10^{24}

(3) 6×10^{23}

(4) 6×10^{24}

88. Ammonia gas is passed into water, yielding a solution of density 0.93 g/cm^3 and containing 18.6% NH_3 by weight. The mass of NH_3 per cc of the solution is

(1) 0.17 g/cm^3

(2) 0.34 g/cm^3

(3) 0.51 g/cm^3

(4) 0.68 g/cm^3

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89. Copper has two naturally occurring isotopes ^{63}Cu and ^{65}Cu. If the average atomic mass of copper is 63.546 u, then natural abundance of lighter isotope of Cu will be

(1) 46.4%

(2) 27.3%

(3) 54.2%

(4) 72.7%

90. Electronic configuration of Cu^{+} is

(1) [Ar] $3d^{10}$

(2) [Ar] $3d^{9}\ 4s^{1}$

(3) [Ar] $3d^{8}\ 4s^{2}$

(4) [Ar] $3d^{7}\ 4s^{1}$

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91. Read the following **assertion (A)** and r**eason (R**) statements and select the correct option.
Assertion (A) : Habitat loss and fragmentation is the most important cause driving animals and plant extinction.
Reason (R) : Anthropogenic extinction is causing a sixth mass extinction of species.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

92. Read the statements (A – D) w.r.t. predators and select the correct option stating which ones are true **(T)** and false **(F)**.
(A) They transfer energy across trophic levels
(B) They keep prey population under control.
(C) They maintain species diversity in a community
(D) Predators in nature are 'Prudent'

	A	B	C	D
(1)	F	T	T	F
(2)	F	T	T	T
(3)	F	T	F	T
(4)	T	T	T	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

93. Major site of decomposition in the soil and water bodies respectively is

(1) Bottom and upper layer

(2) Upper and bottom layer

(3) Upper and top layer

(4) Bottom and top layer

94. The extinction of passenger pigeon was due to

(1) Bird flu virus infection

(2) Non-availability of the food

(3) Over-exploitation by humans

(4) Increased number of predatory birds

95. If we analyse the species-area relationship among the entire continent, we will find that the regression coefficient (z) values in the range of

(1) 1.2 to 2.6

(2) 0.1 to 0.2

(3) 0.6 to 1.2

(4) 0.2 to 0.6

96. Read the following statements and select the **correct** option.
Statement-A : India with 8.1% of the land area possesses 2.4% species diversity of the world.
Statement-B : India is one of the 12 megadiversity countries of the world.

(1) Only A is incorrect

(2) Only B is incorrect

(3) Both A and B are incorrect

(4) Both A and B are correct

97. Which of the following species have become extinct due to overexploitation by human?

(1) Nile perch

(2) *Lantana*

(3) African catfish

(4) Steller's sea cow

98. Passenger pigeon became extinct mainly due to

(1) Alien species invasions

(2) Co-extinctions

(3) Overexploitation

(4) Habitat loss and fragmentation

99. Standing crop is expressed as

(1) The numbers or biomass of organisms per unit area

(2) Total amount of nutrients present in soil at any time

(3) Total number of primary producers at any time in an area

(4) Total amount of energy at any trophic level at any time

100. A population characteristic of a species susceptible to extinction is

(1) Low trophic level in food chain

(2) Inability to switch over to alternate food source

(3) Wide range of distribution

(4) High biotic potential

101. Choose the **correct** type of ecological pyramid on the basis of following data of dry weights.
(i) Secondary consumer – 11 kg/m^2
(ii) Tertiary consumer – 1.5 kg/m^2
(iii) Primary producer – 809 kg/m^2
(iv) Primary consumer – 37 kg/m^2

(1) Upright pyramid of biomass

(2) Inverted pyramid of biomass

(3) Inverted pyramid of energy

(4) Upright pyramid of number

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102. Which one is not a cause of biodiversity loss?

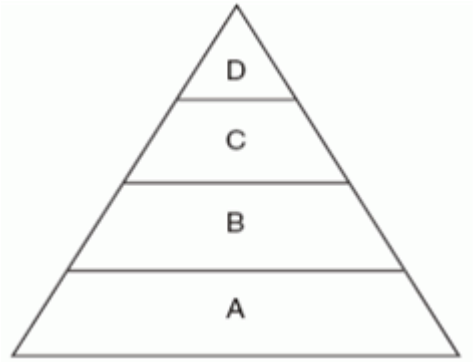
(1) Alien species invasion

(2) Endemism

(3) Overexploitation

(4) Co-Extinction

103. Which statement about the pyramid of energy shown below is correct?



(1) The amount of energy needed to sustain the pyramid enters at level D.

(2) The total amount of energy decreases with each successive feeding level from D to A.

(3) The amount of energy is identical in each level of the pyramid.

(4) The total amount of energy at level D is less than the total amount of energy at level B.

104. In global biodiversity of plants, the

(1) Angiosperms represent maximum number

(2) Algae represent the minimum number

(3) Fungi represent the minimum number

(4) Mosses represent the maximum number

105. The rate of decomposition of detritus will be high when

(1) Temperature is less than 10°C

(2) Air is absent

(3) Moisture is optimum

(4) Detritus contains high amount of chitin

106. In most ecosystems, all the pyramids of number, energy and biomass are upright. It indicates that
(a) Producers are more in number and biomass than the herbivores.
(b) Herbivores are less in number and biomass than the carnivores.
(c) Energy at a lower tropic level is always more than at a higher level.
Choose the **correct** option.

(1) Only (a) is correct

(2) Only (b) is correct

(3) Both (a) and (c) are correct

(4) Both (b) and (c) are correct

107. Read the following statements and choose the **correct** option.
Statement 1: Food webs provide stability to ecosystem.
Statement 2: Food web operates because of taste preference for particular food and unavailability of food.

(1) Only statement (2) is correct

(2) Only statement (1) is correct

(3) Both statements are correct

(4) Both statements are incorrect

108. Which of the following represents largest man made ecosystem?

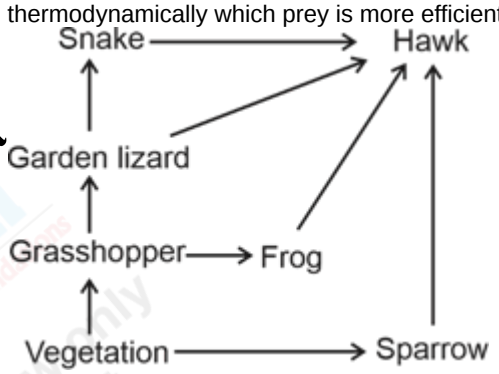
(1) Zoo

(2) Garden

(3) Aquarium

(4) Agroecosystem

109. In the following food web, if 100 kilojoule energy is available in the form of food at producer level, then thermodynamically which prey is more efficient for hawk?



(1) Snake

(2) Frog

(3) Garden lizard

(4) Sparrow

110. Which of the following statements is **not** true w.r.t. hotspots?

(1) Number of endemic species is very high at hotspots

(2) Hotspots have very high level of species richness.

(3) Hotspots are least threatened reservoir of plant and animal life on earth

(4) 34 hotspots are identified globally.

111. Mark the **odd** one w.r.t. functional component of ecosystem

(1) Productivity

(2) Energy flow

(3) Stratification

(4) Decomposition

112. Species diversity

(1) Is diversity of a species at genetic level

(2) Is diversity of species within a region

(3) Play a key role in process of speciation

(4) Enables a population to adapt to its environment

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113. How many organisms given below are primary consumers?
Phytoplanktons, Wolf, Cow, Large fish, Grasshopper

(1) Two

(2) Three

(3) Four

(4) Five

114. The fast dwindling ___A___ forest is estimated to produce ___B___ percent of the total oxygen in earth's atmosphere.

(1) A – Siberian, B – 30

(2) A – Temperate, B – 20

(3) A – Alpine, B – 15

(4) A – Amazon, B –20

115. Consider the following six statements (a-e) about causes of biodiversity losses.
a. Tropical rain forests once covering more than 14% of the earth's land surface, these forests now cover no more than 0.6%.
b. Biological systems should not be exploited beyond the degree of their renewability.
c. Nile perch, a large predator fish was introduced into Lake Victoria of East Australia.
d Lantana is an indigenous plant of our country.
e. Recent illegal introduction of the American catfish for aquaculture purposes is posing a threat to the indigenous catfishes in our rivers.
Which of the above statement(s) is/are correct?

(1) a & b

(2) b, c & d

(3) Only b

(4) b, d & e

116. In aquatic ecosystem, major conduit for energy flow is

(1) Grazing food chain

(2) Detritus food chain

(3) Auxiliary food chain

(4) Parasitic food chain

117. Consider the given assertion and reason, and choose the **correct** option.
Assertion: The pyramid of biomass in sea is generally inverted.
Reason: The biomass of fishes far exceeds that of phytoplankton.

(1) Both the assertion and reason are true and the reason is the correct explanation of assertion.

(2) Both the assertion and reason are true but the reason is not the correct explanation of assertion.

(3) Assertion is true but the reason is false statement.

(4) Both assertion and reason are false statements.

118. Choose the **odd** one w.r.t. significance of mitosis.

(1) It is essential for growth and development of multicellular organisms

(2) It increases the genetic variability in the population of organisms

(3) It is responsible for regeneration and healing

(4) It contributes to the continuous growth of plants throughout their life

119. Mitochondria and chloroplast, both are considered as semi-autonomous cell organelles. How many of the given features is/are exclusive to mitochondria?
(a) Presence of dsDNA
(b) Oxidation of carbohydrates
(c) Presence of ETS
(d) Surrounded by double membrane
(e) ATP synthesis

(1) 2

(2) 3

(3) 1

(4) 4

120. Read the following statements and state **True (T)** or **False (F)**.
A. Decision of a cell to divide occurs in G_1 phase.
B. G_1 to S transition is carried out by maturation promoting factor formed by cdc 2 kinase.
C. Maximum cytoplasmic growth occurs in G_2 phase.
D. Cyclins are proteins that activate protein kinases to regulate prokaryotic cell cycle.

(1) A-(T), B-(T), C-(T), D-(F)

(2) A-(T), B-(F), C-(T), D-(T)

(3) A-(F), B-(T), C-(F), D-(T)

(4) A-(T), B-(F), C-(T), D-(F)

121. Which among the following events take place during meiosis I but not during mitosis?
a. Nuclear membrane disappears
b. Pairing of homologous chromosomes
c. Terminalization of chiasmata
d. Microtubules attach to kinetochore of chromosomes
e. Separation of homologous chromosome
f. Arrangement of chromosomes on equatorial plate
Choose the **correct** ones

(1) a, b and e only

(2) b and d only

(3) b, c and e only

(4) c, d, e and f only

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122. Read the given statements and choose the **correct** option
Statement A: The final stage of prophase I is marked by dissolution of synaptonemal complex and can lasts for years in vertebrates.
Statement B: During mitosis, best stage to study morphology of chromosomes is characterized by all the chromosomes coming to lie at the equator.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

123. Cytokinesis in a plant cell is achieved by the formation of cell plate instead of a cleavage furrow, which is formed in

(1) Centripetal manner

(2) Centrifugal manner

(3) Both centripetal and centrifugal manner

(4) Equational manner

124. Anaphase II can be differentiated from anaphase I as in the former

(1) Homologous chromosomes separate from each other and start moving towards the pole

(2) Chromosomes will appear as tetrad

(3) The centromere holding the two chromatids splits and allows the separation and movement of two chromatids

(4) The two homologous chromosomes are not separated at chiasmata

125. All of the following are significance of equational division, **except**

(1) It helps in maintaining nucleo-cytoplasmic ratio of the cell

(2) It contributes in the cell repair

(3) It helps in increasing the genetic variability in the daughter cells

(4) It helps in repair of upper layer of epidermis, cells of the lining of gut and blood cells

126. All of the following are the roles of mesosomes in prokaryotic cells, **except**

(1) Formation of thin filamentous extensions on outer surface of the cell for motility

(2) DNA replication and distribution to daughter cells

(3) Cell wall formation and surface area expansion of plasma membrane

(4) Respiration and secretion processes

127. Choose the **incorrect** match.

(1) Nucleolus, golgi complex and ER reform –Telophase

(2) Furrow in the plasma membrane – Animal cell

(3) Cell wall formation starts at the periphery and meets in the centre – Plant cell

(4) Restoring nucleo-cytoplasmic ratio – Mitosis

128. Select **odd** one w.r.t. mitochondria.

(1) These are semi-autonomous cell organelle

(2) These are site of aerobic respiration

(3) Matrix possesses linear DNA molecule

(4) They divide by fission

129. Match the following Columns and select the **correct** option.

	Column-I		**Column-II**
a.	70S ribosome	(i)	Contains Carotenoid pigments
b.	Chlorophyll	(ii)	Site for protein synthesis
c.	Chromoplast	(iii)	Storage of starch
d.	Amyloplast	(iv)	Trapping light energy

(1) a(ii), b(iv), c(iii), d(i)

(2) a(ii), b(i), c(iv), d(iii)

(3) a(ii), b(iv), c(i), d(iii)

(4) a(iv), b(i), c(ii), d(iii)

130. Double metaphase plate is formed during

(1) Prophase II

(2) Metaphase I

(3) Anaphase II

(4) Metaphase II

131. Identify the following statements as True (T) or False (F) and choose the **correct** option.
A. The least number of meiotic divisions required to form 100 pollen grains in a typical angiospermic plant is 125.
B. The time required by a human cell in culture, to prepare itself for cell division is more than 22 hours.
C. In all types of eukaryotic cells, vacuoles can occupy 90% of total volume of the cell.
D. 27 microtubules are present in the core of the structure that gives rise to basal body of locomotory structure in a eukaryotic cell.

	A	**B**	**C**	**D**
(1)	T	T	F	F
(2)	F	F	T	T
(3)	F	T	F	T
(4)	T	F	T	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

132. How many of the given cellular components is/are present in the sole members of the kingdom monera?

Mitochondria, Chloroplast, Ribosome, Cell wall, Peroxisome

(1) One

(2) Two

(3) Four

(4) Three

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133. A cell at G_1 phase contains 30 picogram DNA and 20 chromosomes. What will be the number of chromosomes and DNA content respectively after G_2 phase?

(1) 20, 60

(2) 40, 90

(3) 15, 80

(4) 60, 20

134. In some prokaryotes like cyanobacteria, there are membranous extensions into the cytoplasm called_____ which contain photosynthetic pigments.
Fill in the above blank with the **correct** option given below

(1) Mesosomes

(2) Chromatophores

(3) Slime layer

(4) Lysosomes

135. The rate of decomposition will be quicker if

(1) Detritus is rich in lignin

(2) The process occurs in very low temperature

(3) Detritus is rich in nitrogen and sugars

(4) The process occurs in absence of oxygen

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BOT-11

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ZOOLOGY-12

136. The novel strategy used by Eli Lilly in 1983 for production of humulin by genetic engineering was

(1) Production of Pro-insulin by *E.coli* and maturation in bioreactors

(2) Production of A and B chains separately and then combined

(3) Production of chains A and B together in the same host cell and then extracted

(4) Production of mature insulin directly by *E.coli* and requiring no further combining of chains

137. Choose the organism, whose natural product is used as a clot buster for removing thrombus

(1) *Entamoeba coli*

(2) *Streptococcus*

(3) *Methanococcus*

(4) *Vibrio cholerae*

138. A mutated gene can be identified directly by using

(1) Probe

(2) ELISA

(3) Microscopy

(4) Serum analysis

139. A type-II restriction enzyme *Sal* I recognises 6 base pair palindrome as follows:

5′... G▼TCGAC...3′
3′... CAGCT▲G...5′

The *Sal* I cut both strands at the position indicated by the arrows and produces
(a) Cohesive ends
(b) Blunt ends
(c) Flush ends

(1) a only

(2) b only

(3) a and c

(4) b and c

140. The copy number of the linked DNA in the vector, is controlled by a sequence called as

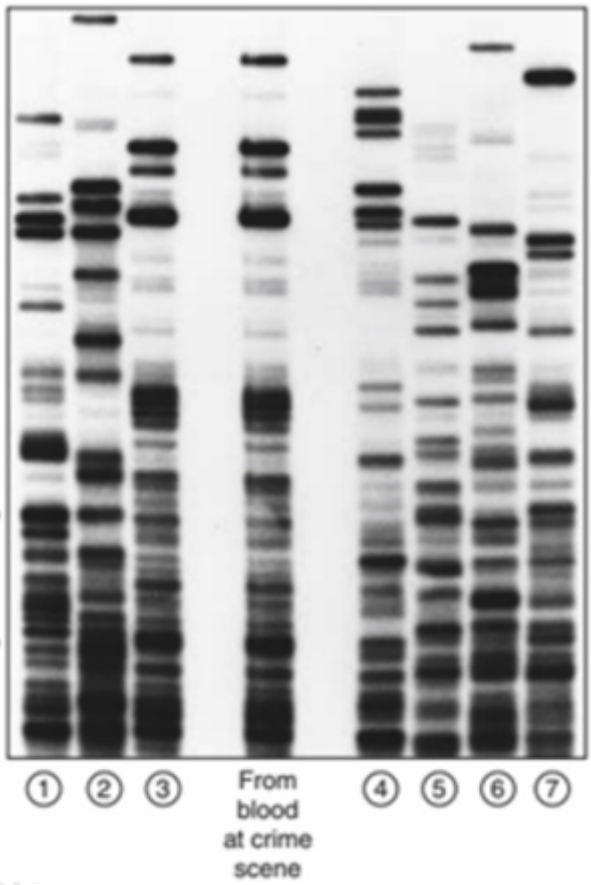
(1) Ori site

(2) Selectable marker

(3) Palindromic sequence

(4) Recognition site

141. DNA fingerprinting relies on PCR amplification and gel electrophoresis to detect molecular markers.
These DNA fingerprints show DNA from a crime scene (in middle), along with the DNA profiles of seven suspects. Pick the suspect whose DNA profile matches with DNA from the crime scene.



Choose the correct option.

(1) 3

(2) 4

(3) 6

(4) 2

142. Entry of DNA in a competent animal cell is due to the use of which ingredient/condition?

(1) Heat shock at 42°C

(2) Cold treatment at 4°C

(3) Cold shock with calcium salts

(4) Lysozyme

143. In gel electrophoresis,

(1) Smaller fragments travel slower than larger fragments

(2) Larger DNA fragments stay closer to cathode as compared to smaller fragments

(3) Ethidium bromide is used to visualise DNA bands in visible light

(4) DNA is extracted from gel by spooling

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144. Strains of *Bacillus thuringiensis* (Bt) are used in producing

(1) Bioinsecticidal plants

(2) Biomineralisation

(3) Biometallurgical techniques

(4) Biofertilizers

145. An enzyme belonging to _____ plays a vital role in the formation of rDNA as it helps in linking the alien DNA and cloning vector and is also called _____.
Select the correct option to fill in the blanks respectively.

(1) Class VI; molecular scissors

(2) Class IV; molecular glue

(3) Class VI; molecular glue

(4) Class IV; molecular scissors

146. Select the method by which DNA is removed with the help of glass rod after precipitation of DNA due to the addition of chilled ethanol.

(1) Spooling

(2) Precipitation

(3) Condensation

(4) Elution

147. _____ produced using RDT is used as a clot buster for removing thrombus. Select the **correct** option which fills the blank

(1) Streptokinase

(2) Clotting factor VIII

(3) Erythropoeitin

(4) Human interleukin

148. RNAi is a naturally occurring mechanism and leads to silencing of genes due to formation of

(1) ds-DNA

(2) ds-RNA

(3) ss-DNA

(4) ss-RNA

149. Presence of coronavirus, if present even in little amount can be traced using

(1) RT-PCR

(2) Urine analysis

(3) Stool analysis

(4) Widal test

150. Nematode *Meloidogyne incognita* cannot survive in transgenic tobacco expressing.

(1) dsDNA

(2) siRNA

(3) ssDNA

(4) sense mRNA

151. Which of the following hormone was extracted earlier from slaughtered cattle but are now produced using biotechnology?

(1) Cortisol

(2) Insulin

(3) Testosterone

(4) Estradiol

152. Given below are two statements : One is labelled as **Assertion** and the other is labelled as **Reason.**
Assertion : Bt toxin does not kill the bacterium *Bacillus thuringiensis*.
Reason : This protein exists as an inactive protoxin in *Bacillus* but gets activated in midgut of insect.
Select the **correct** option.

(1) Both Assertion & Reason are true and reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

153. Bollworm successfully attacks

(1) Bt Tomato

(2) *Bacillus thuringiensis*

(3) Cotton plant

(4) Bt cotton

154. Which method is preferred for gene transfer to overcome the problem of strong cell wall in plant tissue?

(1) Microinjection

(2) Gene gun

(3) Heat shock

(4) Electroporation

155. If a gene encoding the enzyme β-galactosidase is inserted within the selectable marker gene of the plasmid of *E.coli* then, in the presence of a chromogenic substrate

(1) Non-recombinant colonies will produce the blue colour

(2) All transformants will remain white due to inactivation of selectable marker gene

(3) Recombinant colonies will be blue in colour

(4) Non- transformants will appear pink in colour

156. Bioremediation of oil spills is carried out by which genetically engineered microorganism?

(1) *Escherichia coli*

(2) *Rhizobium leguminosarum*

(3) *Salmonella typhi*

(4) *Pseudomonas putida*

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157. Which of the following Indian plants have either been patented or attempts have been made to patent them by western nations for their commercial use?
(a) Daffodil
(b) Neem
(c) Turmeric
Choose the correct option.

(1) Only (a) and (b)

(2) Only (a) and (c)

(3) (a), (b) and (c)

(4) Only (b) and (c)

158. Which of the following is/are direct method of gene transfer?
(A) Electroporation
(B) Microinjection
(C) Biolistic
(D) Disarmed retrovirus mediated vector
Choose the **correct** option.

(1) C and D

(2) B, C and D

(3) A and D

(4) A, B and C

159. Match the column I and column II and choose the **correct** option.

	Column I		**Column II**
a.	*cry IAc*	(i)	Tracy
b.	*cry IAb*	(ii)	Cotton bollworms
c.	α-lactalbumin	(iii)	Rosie
d.	α-1-antitrypsin	(iv)	Corn borer

(1) a(ii), b(iv), c(i), d(iii)

(2) a(iv), b(ii), c(i), d(iii)

(3) a(ii), b(iv), c(iii), d(i)

(4) a(i), b(ii), c(iii), d(iv)

160. Large scale production of genetically identical plants in a short time through tissue culture is possible because

(1) Somaclonal variation increases genetic diversity

(2) All regenerated plants arise from meiotically dividing cells of a single parent

(3) Plant cells are totipotent and in controlled nutritional and environmental conditions can grow and multiply continuously

(4) Tissue culture is a form of sexual reproduction that leads to the birth of a new generation

161. DNA fragments are resolved based on their size in agarose gel electrophoresis through sieving effect. The movement of DNA fragments is based on the property of negative charge on

(1) Sugar moieties

(2) Nitrogenous bases

(3) Phosphate moieties

(4) Nucleoside

162. Read the following statements.
Statement A: Genetically modified crop is used to maximise the use of fertilizers and chemicals so that their harmful effects on the environment are reduced.
Statement B: Genetic modification has been used to create tailor-made plants to supply alternative resources to industries , in the form of starches, fuel and pharmaceuticals.
Select the **correct** option.

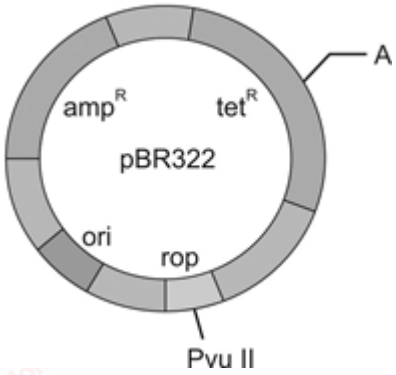
(1) Only statement B is correct

(2) Only statement A is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

163. Restriction enzyme whose restriction site is represented by the region marked 'A' is



(1) *Pvu* I

(2) *Bam*H I

(3) *Pst* I

(4) *Hin*d III

164. Genetically modified crops show all of the following properties, **except**

(1) More tolerance to abiotic stresses

(2) Reduced reliance on chemical pesticides

(3) Reduced post harvest losses

(4) Reduced efficiency of mineral usage by plants

165. A 'selectable marker' helps in identification of

(1) Type of vector

(2) Copy number of plasmid

(3) Transformants and non transformants

(4) Replication pattern of vector

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166. Match the following.

Column I		Column II	
(a)	Setae	(i)	Scattered over all the segments except first two
(b)	Cuticle	(ii)	Present in the middle of the segment
(c)	Nephridiopore	(iii)	Ventrolateral side of the intersegmental groove
(d)	Spermathecal aperture	(iv)	Secreted by epidermis and covers the body

(1) a(ii), b(iv), c(i), d(iii)

(2) a(iii), b(ii), c(iv), d(i)

(3) a(ii), b(i), c(iv), d(iii)

(4) a(i), b(iii), c(ii), d(i)

167. Single layer of flattened polygonal cells with irregular boundaries form

(1) Squamous epithelium

(2) Cuboidal epithelium

(3) Columnar epithelium

(4) Ciliated columnar epithelium

168. How many of the structures mentioned in the box below are included in the broad class of tissues which is most abundant and widely distributed in the body of complex animals?

Cartilage, Aerolar tissue, Blood, Adipose tissue, Bone

Choose the **correct** option.

(1) Two

(2) Three

(3) Four

(4) Five

169. Read the given statements.
Statement A. Skeletal and cardiac muscle fibres are striated in appearance.
Statement B. These cells have transverse stripes when viewed under the microscope.
Select the **correct** option.

(1) Statement A is correct and B is the correct explanation of statement A

(2) Statement A is correct but B is not the correct explanation of statement A

(3) Statement A is incorrect but B is correct

(4) Both statements A & B are incorrect

170. Choose the **incorrect** statement w.r.t cell junctions.

(1) They can serve as structural and functional links between adjacent cells

(2) Tight junctions prevent leakage across a tissue

(3) Adhering junctions primarily facilitate communication between cells

(4) Gap junctions connect cytoplasm of adjoining cells

171. Select the **incorrect** option w.r.t. compound epithelium.

(1) Rests on a vascular basement membrane

(2) Has superficial layers of squamous cells at some sites

(3) Present in the inner lining of pancreatic ducts

(4) Covers the dry surface of the skin

172. In epithelial tissue

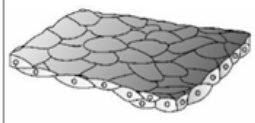
(1) Cells are compactly packed with large amounts of intercellular matrix

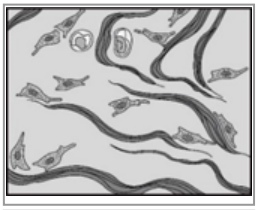
(2) A free surface is present which faces either a body fluid or the outside environment

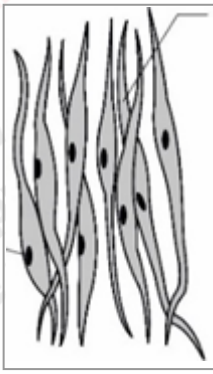
(3) The tissue provides a protective covering or lining for all parts of the body

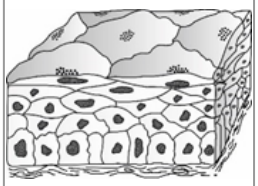
(4) The cells are always loosely packed with little intercellular matrix

173. The four sketches (A, B, C and D) given below, represent four different types of animal tissues. Which one of these is **correctly** identified in the options given, along with its correct location and function?

(A) 

(B) 

(C) 

(D) 

		Tissue	Location	Function
(1)	(A)	Cuboidal epithelium	Wall of blood vessels	Absorption
(2)	(B)	Compound epithelium	Dry surface of skin	Provide protein against chemical and mechanical stresses
(3)	(C)	Dense regular connective tissue	Skin	Connects bone to bone
(4)	(D)	Striated muscle fibres	Heart	Help in bringing about involuntary movement

(1) (1)

(2) (2)

(3) (3)

(4) (4)

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174. Choose the odd one w.r.t. compound epithelium

(1) It consists of more than one layer of cells

(2) Only the cells of deepest layer rest on basement membrane

(3) They do not provide protection to underlying tissues against mechanical stresses.

(4) It has limited role in secretion and absorption of substances

175. Read the following statements w.r.t. epithelial tissue and choose the **correct** option.
Statement A – Basal lamina is composed of mucopolysaccharides and glycoproteins.
Statement B – Reticular lamina is composed of collagen fibres of the underlying connective tissue.

(1) Both statement are incorrect.

(2) Statement B is incorrect.

(3) Only statement A is correct.

(4) Both statements are correct.

176. There are two kinds of cells in the nervous system:
I : ___A___, which receive and transmit information to other cells.
II : ___B___, which do not transmit information *i.e,* are non excitable.
Choose the option which fill the blanks **correctly**

(1) **A –** Neuroglial, **B** – Neurons

(2) **A –** Neurons, **B** – Neuroglial

(3) **A –** Corpuscles, **B** – Neurons

(4) **A –** Fibres, **B** – Neurons

177. Which of the following cartilage is present at the tip of our nose?

(1) Hyaline cartilage

(2) Elastic cartilage

(3) White fibrous cartilage

(4) Calcified cartilage

178. Select the **correct** option w.r.t. dense irregular connective tissue.

(1) Attaches skeletal muscles to bones

(2) Contains osteocytes, macrophages and fibroblasts

(3) Cells present do not secrete modified polysaccharides to form its matrix

(4) Present in the skin

179. Which of the following does **not** show presence of connective tissue fibres of collagen and elastin?

(1) Bone

(2) Blood

(3) Cartilage

(4) Tendon

180. Complete the analogy by selecting **correct** option.
Ciliated epithelium : Fallopian tubes :: _____ : Alveoli.

(1) Columnar epithelium

(2) Muscular tissue

(3) Squamous epithelium

(4) Neural tissue

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